DOOB'S PROOF OF MLE CONSISTENCY

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Abstract. A short proof of the consistency of Fisher's maximum likelihood estimator, following Doob's proof, is presented.

Introduction

The consistency of Fisher's maximum likelihood estimator (MLE) [10] is a basic result discovered a century ago. This result is typically subsumed under the rubric of M-estimators, and derived via standard techniques dating back to Cramér [6] and Wald [23]. For a more recent treatment, van der Vaart [22] is standard.

In teaching this material, I found Doob's consistency proof [9] for the MLE more motivated and accessible to students, especially to students already exposed to information and entropy as part of a data science or machine learning course. The fact that the proof covers general parameter θ spaces and general state X spaces with no extra effort only adds to its appeal.

The aim of this note is to present this proof, together with some historical terminology remarks, in a self-contained manner. There is no novelty here except, perhaps, the conciseness of the presentation. By using the law of large numbers in its strong form, we derive MLE consistency in its strong form.

We start with the setup and assumptions, then state and prove consistency. After that, we attempt to give a coherent historical account of entropy terminology, which is an interesting story in itself, then we end by showing the assumptions hold for the multivariate normal case.

Surprisal and Information

Suppose a random variable X is sampled repeatedly, resulting in i.i.d. copies X_1, X_2, $\dots$ of X. Suppose the density $p(x, \theta^*)$ of X is one of a parametric family of densities $p(x, \theta)$. Then θ^* is the *true parameter*.

More explicitly, let E denote the background expectation. Then the expectation corresponding to the parameter θ is

$$E_\theta(f(X)) = E(p(X, \theta) f(X)),$$

and the true expectation is

$$E_*(f(X)) = E(p(X, \theta^*) f(X)).$$

With this setup, the goal is to estimate or recover θ^* from X_1, X_2, $\dots$.

Below, "with probability one" always refers to the true probability P_* corresponding to the true expectation E_*. The background expectation E plays no role by itself, as it enters only through E_*. Because of this, by replacing $p(X, \theta)$ by $p(X, \theta)/p(X, \theta^*)$, our convention will be $E = E_*$, hence $p(X, \theta^*) \equiv 1$.

Crucial to the analysis is the *surprisal* [17, 20]

$$S(X,\theta) = \log(1/p(X,\theta)). \tag{1}$$

This terminology is intuitive: Low probability equals high surprisal. By our convention, $S(X,\theta^*) \equiv 0$. Using the surprisal, the *relative information* is

$$I(\theta^*,\theta) = E_*(S(X,\theta)). \tag{2}$$

This quantity appears in many fields, in different forms, and is often called relative entropy or Kullback-Leibler divergence. Because of the widely differing terminology, below we attempt to give a coherent account of the terminology history of I.

Whatever it is called, the fundamental property of $I(\theta^*,\theta)$ is its ability to isolate the true density: $I(\theta^*,\theta) \geq 0$, and $I(\theta^*,\theta) = 0$ iff $p(x,\theta) \equiv 1$. In general, $I(\theta^*,\theta)$ is defined differently, and may be infinite, but is finite iff $S(X,\theta)$ is P_* integrable, in which case (2) holds [7, 8].

The simplest example is a categorical random variable X distributed according to $\theta^* = (\theta_1^*, \theta_2^*, \dots, \theta_d^*)$. If the values of X are 1, 2, …, d, the mass function of X is $p(X,\theta^*) = \theta_X^*$. If $\theta = (\theta_1, \theta_2, \dots, \theta_d)$ is any other distribution, the surprisal is $S(X,\theta) = \log(1/\theta_X)$, and

$$I(\theta^*,\theta) = \sum_{i=1}^{d} \theta_i^* \log(\theta_i^*/\theta_i). \tag{3}$$

To match (2) and (3), one must reduce to $E = E_*$, replacing θ by θ/θ^*.

To achieve consistency, there are three natural assumptions, the first of which is

A. *(Identifiability):* $\theta \neq \theta'$ implies $p(x,\theta) \neq p(x,\theta')$, hence $I(\theta^*,\theta) > 0$.

Without identifiability, one cannot hope to extract θ^* from $p(x,\theta^*)$.

The *likelihood* and *mean surprisal* are

$$L_n(\theta) = \prod_{k=1}^{n} p(X_k,\theta), \qquad \bar{S}_n(\theta) = \frac{1}{n}\log(1/L_n(\theta)) = \frac{1}{n}\sum_{k=1}^{n} S(X_k,\theta).$$

The mean surprisal is also called the negative-log-likelihood.

By the law of large numbers, the mean surprisal $\bar{S}_n(\theta)$ converges to $E_*(S(X,\theta))$ as $n \to \infty$, with probability one. The true parameter θ^* minimizes the expected surprisal $E_*(S(X,\theta))$. Based on this, it is natural to estimate θ^* by the minimizer of $\bar{S}_n(\theta)$ or, equivalently, the maximizer of $L_n(\theta)$: For each $n \geq 1$, a *maximum likelihood estimator (MLE)*, if it exists, is a random variable $\hat{\theta}_n = \hat{\theta}_n(X_1, X_2, \dots, X_n)$ maximizing $L_n(\theta)$.

A function $f(\theta)$ is *proper* if its sublevel sets $f(\theta) \leq c$ are compact subsets of the parameter space, for every level c. A well-known elementary fact is the existence of minimizers of continuous proper functions. Since $\hat{\theta}_n$ should minimize $\bar{S}_n(\theta)$, this leads us to the second assumption

B. *(Properness):* There is a continuous proper $f(\theta)$ such that, with probability one, the inequality

$$f(\theta) \leq \bar{S}_n(\theta), \qquad \text{for all } \theta, \tag{4}$$

holds for all but finitely many n. Then **B** implies, with probability one, an MLE $\hat{\theta}_n$ exists for all but finitely many n.

We will need the convergence $\bar{S}_n(\theta) \to E_*(S(X,\theta))$ to be simultaneous across all θ, in the sense

$$\bar{S}_n(\theta) \to E_*(S(X,\theta)), \qquad \text{as } n \to \infty, \qquad \text{for all } \theta, \tag{5}$$

with probability one. For this we need the third assumption

C. *(Continuity):* The parameter space is separable and there is a continuous $f(\theta,\theta')$ and integrable $g(x)$ satisfying

$$|S(X,\theta) - S(X,\theta')| \le f(\theta,\theta')g(X) \qquad \text{and} \qquad f(\theta,\theta) = 0.$$

Then **C** and a separability argument establishes the validity of (5) with probability one. Combining (4) and (5), **B** and **C** together imply the properness of $E_*(S(X,\theta))$. Because of this, before establishing **A**, **B**, **C**, it is best to first establish properness of $I(\theta^*,\theta)$ as a reality check.

The naturality of **A**, **B**, **C** is illustrated by the multivariate normal example, worked out below.

Consistency

Theorem. *Assume a population has a distribution depending on a parameter θ, and let θ^* be the population's true parameter. Assume also the relative information $I(\theta^*,\theta)$ is finite for all θ. Then, under* **A**, **B**, **C**, *with probability one, an MLE sequence $\hat{\theta}_n$ exists, for all but finitely many n, and any such sequence converges to θ^* as $n \to \infty$.*

By the law of large numbers,

$$\frac{1}{n}\sum_{k=1}^{n} g(X_k) \to E_*(g(X)), \qquad \text{as } n \to \infty, \tag{6}$$

with probability one. Fix an outcome for which (4), (5), (6) hold, except, for (4), for finitely many n. We show $\hat{\theta}_n \to \theta^*$ as $n \to \infty$ for this outcome.

By **B**,

$$f(\hat{\theta}_n) \le \bar{S}_n(\hat{\theta}_n) \le \bar{S}_n(\theta^*), \qquad \text{for almost all } n.$$

By (5), the right side is bounded as $n \to \infty$. Since $f(\theta)$ is proper, the sequence $\hat{\theta}_n$ lies in a compact set in the parameter space.

By **C**,

$$\bar{S}_n(\theta) - \bar{S}_n(\theta^*) \le \bar{S}_n(\theta) - \bar{S}_n(\hat{\theta}_n) \le f(\theta,\hat{\theta}_n) \cdot \frac{1}{n}\sum_{k=1}^{n} g(X_k).$$

Now suppose θ is a limit point of the sequence $\hat{\theta}_n$, so $\hat{\theta}_n \to \theta$ along a subsequence. Then $f(\theta,\hat{\theta}_n) \to 0$ along a subsequence, hence

$$I(\theta^*,\theta) = E_*(S(X,\theta)) = E_*(S(X,\theta) - S(X,\theta^*)) \le 0.$$

But $I(\theta^*,\theta) \ge 0$, hence $I(\theta^*,\theta) = 0$. From **A**, this implies $\theta = \theta^*$, establishing the result.

Terminology History

The relative information $I(\theta^*, \theta)$ is often called "relative entropy" [5, 14]. The term "entropy", originating in thermodynamics, was coined by Clausius in his 1865 paper [4]. Boltzmann, in his 1872 paper [2], derived his H theorem, using a quantity H proportional to the negative of entropy. Gibbs, in his 1902 book [11], building on Boltzmann's 1877 combinatorial interpretation [3] and Planck's 1901 paper [16], flipped Boltzmann's sign convention, writing the entropy H as done today.

Shannon, in his 1948 paper [19], independently defined

$$H(p) = -\sum_{i=1}^{d} p_i \log(p_i).$$

Von Neumann had earlier defined quantum entropy in his 1932 book [15], and when Shannon consulted him around 1939–1940 about terminology for H, von Neumann suggested "entropy", which Shannon adopted [21]. However, in his remark "H is then, for example, the H in Boltzmann's H theorem", Shannon conflates his H with Boltzmann's H.

The relative information $I(\theta^*, \theta)$ is jointly convex in (θ^*, θ), a fact that is apparent in the special case (3), whereas $H(p)$ above is strictly concave in p. Because of this, calling both quantities "entropy" creates a mismatch. Indeed, based on the curvature, it is natural to call the *negative* of (3) the *relative entropy*, as information and entropy are opposites. The mismatch is also in Python: `entropy(p)` returns the concave H, but `entropy(p,q)` returns the convex I.

This error is a common psychological error: When measuring the temperature, are we measuring how cold it is, or how hot it is? Of course we are doing both, but this doesn't mean cold equals hot. In fact, cold equals minus hot.[1]

While many authors' terminology conflicts with ours, some authors align partially with our usage. Schervish, in his 1995 text [18], uses "Kullback-Leibler information" for I, which is more descriptive than the widely used but opaque "Kullback-Leibler divergence" [1]. Kullback himself, in his 1959 monograph [13], refers to I as an "information function". Finally, Donsker-Varadhan, in their 1975 large deviations paper [8], denote the rate function as I rather than H, derive its properties in full generality, yet give it no name.

Regarding Doob's 1934 proof, this was at the dawn of abstract probability theory, within the milieu developed by Birkhoff, Daniell, Doob, Hopf, Khintchine, Kolmogorov, Lévy, Radon, and Wiener, and within a year of Kolmogorov's 1933 foundational extension theorem [12].

Multivariate Normal

It is completely straightforward to verify the assumptions for the scalar normal distribution. For the multivariate normal distribution, the situation is nontrivial, due to the matricial complications, so this we do carefully. Here the density is specified by its mean μ^* in $\mathbf{R}^d$ and variance $Q^* > 0$, a positive definite symmetric $d \times d$ matrix, and the surprisal is

$$S(X, \theta) = S(X, \mu, Q) = \frac{1}{2} \log \det(2\pi Q) + \frac{1}{2}(X - \mu) \cdot Q^{-1}(X - \mu). \tag{7}$$

[1] Chinese/Indian/Arab mathematicians figured out negative integers a while ago.

Here $v \cdot w$ is the dot product in $\mathbf{R}^d$. We verify **A**, **B**, **C** for this $S(X,\theta)$.

To carry out the computations, we recall matrix terminology and results. Given symmetric matrices Q and Q', we say $Q \geq 0$ if $v \cdot Qv \geq 0$ for all v, and $Q \geq Q'$ if $Q - Q' \geq 0$. Given vectors v, w in $\mathbf{R}^d$, let $v \otimes w$ be the symmetric matrix whose ij-th entry is $(v_i w_j + v_j w_i)/2$ (this is the *symmetric* tensor product). Then

$$v \otimes w = w \otimes v, \quad v \cdot Qw = \operatorname{trace}(Q(v \otimes w)), \quad 2v \otimes w \leq \epsilon v \otimes v + \frac{1}{\epsilon} w \otimes w, \quad \epsilon > 0.$$

By the eigenvalue decomposition, every symmetric matrix is a linear combination of matrices of the form $v \otimes v$. The determinant $\det(Q)$ is the product of the eigenvalues of Q. With Q invertible symmetric and t a scalar,

$$(Q + tv \otimes v)^{-1} = Q^{-1} - \frac{t}{1 + tv \cdot Q^{-1} v} \cdot (Q^{-1}v) \otimes (Q^{-1}v)$$

and

$$\det(Q + tv \otimes v) = \det(Q) \left(1 + tv \cdot Q^{-1} v\right).$$

These are the matrix facts used below.

By the law of large numbers, with probability one, we have

$$\frac{1}{n} \sum_{k=1}^{n} (X_k - \mu^*) \otimes (X_k - \mu^*) \geq (1 - \epsilon) Q^*$$

for all but finitely many n, and

$$\left(\mu^* - \frac{1}{n} \sum_{k=1}^{n} X_k\right) \otimes \left(\mu^* - \frac{1}{n} \sum_{k=1}^{n} X_k\right) \leq \epsilon^2 Q^*,$$

for all but finitely many n. These are also used below.

The computations are stated as a series of exercises. For $0 < \epsilon < 1/2$, define

$$h_\epsilon(\mu) = (1 - 2\epsilon) Q^* + (1 - \epsilon)(\mu - \mu^*) \otimes (\mu - \mu^*).$$

Then $h_0(\mu) = Q^* + (\mu - \mu^*) \otimes (\mu - \mu^*)$.

Exercise 1. $E_*(S(X,\theta))$ equals one half of

$$\log \det(2\pi Q) + \operatorname{trace}(Q^{-1} h_0(\mu)).$$

Exercise 2. Show $S(X,\theta) = S(X,\theta')$ implies $\theta = \theta'$, verifying **A**.

Exercise 3. $h_\epsilon(\mu)$ is proper in the sense the sublevel sets $h_\epsilon(\mu) \leq cI$ are compact. Moreover if $\delta > 0$ is the lowest eigenvalue of $(1 - 2\epsilon)Q^*$, then $h_\epsilon(\mu) \geq \delta I$.

Exercise 4. If $h(\mu)$ is any proper symmetric matrix-valued function of μ satisfying $h(\mu) \geq \delta I$ with $\delta > 0$, then

$$f(\mu, Q) = \log \det(2\pi Q) + \operatorname{trace}(Q^{-1} h(\mu))$$

is a proper function of (μ, Q). This is the core exercise; do the scalar case first.

Exercise 5. $E_*(S(X, \mu, Q))$ is proper.

Exercise 6. With probability one,

$$\frac{1}{n} \sum_{k=1}^{n} (X_k - \mu) \otimes (X_k - \mu) \geq h_\epsilon(\mu), \qquad \text{for all } \mu,$$

for all but finitely many n.

Exercise 7. Verify **B** with

$$f(\theta) = \frac{1}{2}\log\det(2\pi Q) + \frac{1}{2}\mathrm{trace}(Q^{-1}h_\epsilon(\mu)).$$

Exercise 8. Verify **C**.

Exercise 9. By differentiating $\bar{S}_n(\mu + t\nu, Q)$ and $\bar{S}_n(\mu, Q + tv \otimes v)$ at $t = 0$, show

$$\hat{\mu}_n = \frac{1}{n}\sum_{k=1}^n X_k, \qquad \hat{Q}_n = \frac{1}{n}\sum_{k=1}^n (X_k - \hat{\mu}_n) \otimes (X_k - \hat{\mu}_n).$$

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